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Capstone Report: Digestate Management for Biogas-ZE

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Abstract

Digestate management is a critical component of sustainable anaerobic digestion systems. This report presents a comprehensive design strategy for Biogas-ZE that transforms whiskey distillery waste into valuable outputs through efficient solid-liquid separation, nutrient extraction, and valorization of both liquid and solid fractions. A decanter centrifuge was identified as the optimal solution for primary separation due to its high efficiency, compatibility with downstream processes, and alignment with circular economy principles. For the solid fraction, vacuum evaporation was selected as the preferred processing method, offering significant energy savings through heat integration and producing a stable, nutrient-rich output. The final product is suited for land application as a soil amendment, fulfilling regulatory requirements and supporting long-term agricultural sustainability. This project offers an integrated and scalable solution that aligns with Biogas-ZE's zero-waste and low-emission mandate while contributing to broader goals of nutrient recovery, waste reduction, and environmental stewardship.

Contents

1	Introduction	5
2	Goals	5
2.1	Alignment with Biogas-ZE's Sustainability Ethos	5
2.2	Development of Marketable Products	5
2.3	Efficient Nutrient Recovery and Management	5
3	Motivations	6
4	Liquid-Solid Separation Technologies for Digestate Processing	6
4.1	Context and Requirements	6
4.2	Technology Evaluation Framework	6
4.2.1	Membrane Filtration Systems	6
4.2.2	Mechanical Press Systems	7
4.3	Decanter Centrifuge Solution	7
4.3.1	Technical Advantages	7
4.3.2	Operational and Economic Benefits	7
4.4	Alignment with Circular Economy Principles	8
4.5	Conclusion	8
5	Liquid Fraction Processing	9
5.1	Biogas Production During Anaerobic Digestion of Algae	9
5.1.1	Macroalgae	9
5.1.2	Microalgae	9
5.2	Liquid Fraction Filtration Technologies	9
5.2.1	Centrifugal Filtration	10
5.2.2	Reverse and Forward Osmosis	10
5.2.3	Ultrafiltration	10
6	Nutrient Extraction: Struvite	11
6.1	MAPs (Struvite)	11
6.2	Extraction Methods and Technology	11
6.3	Reactor Types for Struvite Formation	12
6.3.1	Stirred Tank Reactors	12
6.3.2	Fluidized Bed Reactors	12
6.4	Final Processing	12
7	Solid Fraction Considerations	13
7.1	Characteristics of Digestate Solids	13
7.2	Key Objectives in Processing Digestate Solids	13
7.3	Key Selection Criteria for Processing Methods	14
7.3.1	Environmental Impact	14
7.3.2	Energy and Efficiency Requirements	14
7.3.3	Economic Viability	14
8	Solids Processing Methods	14
8.1	Low-Processing Methods	14

8.1.1	Direct Land Application	14
8.1.2	Composting	15
8.2	Thermal Conversion Methods	15
8.2.1	Direct Combustion	15
8.2.2	Pyrolysis	15
8.2.3	Hydrothermal Carbonization (HTC)	15
8.3	Drying and Concentration Methods	16
8.3.1	Conventional Thermal Drying	16
8.3.2	Biodrying	16
8.3.3	Vacuum Evaporation	16
9	Recommended Method: Vacuum Evaporation	17
9.1	Key Trade-offs in Method Selection	17
9.2	Advantages Over Alternative Methods	17
9.3	Alignment with Key Selection Criteria	17
9.4	Implementation Considerations	17
10	Disposal of the Dried Solid Product	18
10.1	Key Objectives in Disposing of Dried Digestate Solids	18
10.2	Combustion	18
10.3	Land Application	19
10.3.1	Nutrient Management	19
10.3.2	Addressing Risks	20
10.3.3	Storage and Transportation	20
11	Conclusion	21
A	Figures	i

List of Figures

1	Schematic representation of decanter centrifuge operation showing principal components and material flows.	8
2	Stirred tank reactor schematic showing the pathways for influent, effluent, struvite and reagents	13
3	Fluidized bed reactor schematic showing the pathways for influent, effluent, struvite and reagents	13
4	Overview of a generic vacuum evaporation system	17
5	Proposed digestate management processes	i
6	Process decision flowchart containing all processes considered as part of this project . . .	ii
7	Decision matrix created for choosing the appropriate process for the solids processing . . .	iii

List of Abbreviations

- AD** Anaerobic Digestion
- CF** Centrifugal Filtration
- FBR** Fluidized Bed Reactor
- FO** Forward Osmosis
- HTC** Hydrothermal Carbonization
- MAP** Magnesium Ammonium Phosphate (Struvite)
- NMP** Nutrient Management Plan
- PAH** Polycyclic Aromatic Hydrocarbon
- PET** Polyethylene Terephthalate
- PVC** Polyvinyl Chloride
- RO** Reverse Osmosis
- STR** Stirred Tank Reactor
- UF** Ultrafiltration

1 Introduction

The anaerobic digestion of whiskey distillery waste presents a promising pathway for sustainable energy production and circular waste valorization. However, the resulting digestate poses significant challenges for facilities seeking to maximize resource recovery while minimizing environmental impacts. This project addresses these challenges through the development of a comprehensive digestate management system for Biogas-ZE, a facility committed to zero-emission operations and circular economy principles. The proposed system integrates three primary technological components optimized for digestate characteristics specific to whiskey distillery waste. For solid-liquid separation, a decanter centrifuge system provides superior separation efficiency (85-92%) without chemical additives while maintaining operational reliability with minimal maintenance requirements. The liquid fraction undergoes ultrafiltration and nutrient recovery processes, including algae cultivation for carbon sequestration and struvite crystallization for fertilizer production. Solid fraction management employs vacuum evaporation technology, which achieves effective moisture reduction while preserving nutrient content and supporting efficient energy utilization through waste heat recovery. This report presents a systematic evaluation of available technologies, selection criteria, and design considerations that informed the final system configuration. Performance metrics including separation efficiency, energy consumption, maintenance requirements, and product quality were assessed to identify optimal solutions aligned with Biogas-ZE's operational constraints and sustainability objectives. The integrated system demonstrates how digestate can be transformed from a disposal challenge into valuable agricultural inputs, establishing a model for sustainable resource management in the biogas sector. A full flowchart depicting all choices is presented in Fig. 5.

2 Goals

The primary objective of this project is to design a digestate management system for Biogas-ZE that

transforms waste streams into valuable resources while minimizing environmental impacts. This goal encompasses three interconnected dimensions: sustainability alignment, product development, and nutrient recovery optimization.

2.1 Alignment with Biogas-ZE's Sustainability Ethos

The system prioritizes environmental sustainability through multiple mechanisms. Energy efficiency is achieved through the implementation of technologies such as decanter centrifugation and vacuum evaporation, which optimize energy consumption relative to processing capacity. Chemical input minimization is pursued through the selection of mechanical and thermal processes that operate without flocculants or treatment additives. Circularity is enhanced through the integration of closed-loop systems that maximize the recovery and beneficial reuse of water, nutrients, and organic matter.

2.2 Development of Marketable Products

Commercial viability represents a critical consideration in system design. The technologies are configured to generate standardized outputs with established market applications, including high-purity struvite crystals suitable for fertilizer production and stabilized organic amendments for agricultural use. Economic feasibility is pursued through the implementation of energy-efficient processes and thermal integration with existing facility operations. All products are designed to meet applicable regulatory standards for their intended applications, mitigating compliance risks and enhancing market acceptance.

2.3 Efficient Nutrient Recovery and Management

Nutrient recovery optimization focuses on maximizing the extraction and valorization of key compounds. For struvite ($\text{MgNH}_4\text{PO}_4 \cdot 6\text{H}_2\text{O}$) recovery, the system incorporates precisely controlled crystallization processes with optimized pH management and reagent dosing to achieve high recovery

efficiency. To prevent environmental losses from land-applied materials, the system design includes recommendations for nutrient management plans (NMPs) that align application rates with crop requirements and soil characteristics.

3 Motivations

Digestate management represents both a critical challenge and a strategic opportunity for biogas facilities operating within a circular economy framework. The anaerobic digestion process at Biogas-ZE generates nutrient-rich digestate which, without proper management, can contribute to eutrophication, greenhouse gas emissions, and soil contamination. However, with appropriate treatment, these byproducts can be transformed into valuable outputs including struvite fertilizer, reclaimed water, and stabilized soil amendments. Several factors drive the development of advanced digestate management solutions for whiskey distillery waste. Regulatory compliance represents a primary consideration, as increasingly stringent standards regarding nutrient management, emission control, and waste disposal necessitate more sophisticated treatment approaches. Economic incentives play a significant role, with the potential recovery and sale of valorized products such as struvite and organic amendments offering opportunities to offset operational costs. Environmental sustainability serves as a foundational principle, as Biogas-ZE's zero-emission mandate requires solutions that minimize carbon footprint, energy consumption, and resource utilization throughout the process chain. The development of this comprehensive management system addresses technological gaps in digestate valorization while providing practical solutions tailored to Biogas-ZE's specific operational parameters. Through this approach, the project aims to establish a benchmark for sustainable digestate management within the biogas industry, demonstrating how challenges in waste processing can be transformed into opportunities for resource recovery.

4 Liquid-Solid Separation Technologies for Digestate Processing

4.1 Context and Requirements

The valorization of whiskey distillery waste through anaerobic digestion produces a nutrient-rich digestate that requires efficient separation of its solid and liquid fractions. This separation process serves multiple critical functions including volume reduction for improved handling, facilitation of downstream processing, and enabling the recovery of valuable nutrients through pathways such as microalgae cultivation and struvite precipitation. For Biogas-ZE's zero-emission digestion system, the separation technology needed to satisfy three key criteria: operational efficiency, economic viability, and compatibility with an automated production line requiring minimal maintenance. These requirements guided the comprehensive technology evaluation process to identify the optimal solution for this specific application.

4.2 Technology Evaluation Framework

A systematic evaluation of separation technologies was conducted, comparing performance characteristics against the project's technical and operational requirements. The assessment considered four principal aspects: separation efficiency as measured by solids capture rate and liquid clarity, operational parameters including energy consumption and throughput capacity, maintenance requirements and system reliability, and finally compatibility with downstream valorization processes. This multi-criteria analysis ensured the selected technology would meet both immediate processing needs and long-term sustainability goals.

4.2.1 Membrane Filtration Systems

Membrane technologies, particularly reverse osmosis (RO), demonstrated excellent separation capabilities by forcing digestate through semipermeable membranes under high pressure. RO systems effectively remove both suspended and dissolved solids,

achieving high-purity liquid fractions [1]. However, their application for whiskey distillery digestate presents significant challenges including prohibitive energy consumption typically ranging from 3-10 kWh/m³ and severe membrane fouling that necessitates frequent cleaning and replacement [2]. These operational limitations make RO systems economically unviable for continuous industrial-scale digestate processing.

Forward osmosis (FO) emerged as a potential low-energy alternative, utilizing osmotic pressure gradients rather than mechanical pressure. While FO systems consume 30-50% less energy than RO [3], they introduce operational complexities in draw solution management and show inconsistent performance with variable feed compositions [4]. The technical challenges associated with maintaining stable osmotic gradients and regenerating draw solutions render FO systems suboptimal for continuous, large-scale digestate treatment applications despite their theoretical energy advantages.

4.2.2 Mechanical Press Systems

Mechanical separation technologies were evaluated for their robustness and lower energy requirements compared to membrane systems. The screw press configuration operates by mechanically compressing digestate through a cylindrical screen via a rotating auger, offering advantages in energy efficiency with typical consumption of 0.5-1.5 kWh/m³ and mechanical simplicity [5]. However, performance limitations become apparent when processing whiskey distillery digestate, as screw presses demonstrate reduced efficiency with highly fluid, fine-particle wastes where particle sizes measure below 0.5 mm without pre-thickening [5]. This characteristic makes them less suitable for direct processing of the digestate stream.

Belt press systems employ a series of permeable belts and rollers to gradually dewater solids, capable of processing larger volumes up to 20 m³/h with moderate energy input of 1-2 kWh/m³ [2]. These systems exhibit several drawbacks including susceptibility to clogging with fibrous materials and the requirement for chemical flocculants typically dosed at 2-5 g/kg dry matter. The need for intensive pretreatment and chemical additives conflicts

with Biogas-ZE's commitment to chemical-free processing and minimal pretreatment requirements, despite the apparent throughput advantages of belt press systems.

4.3 Decanter Centrifuge Solution

4.3.1 Technical Advantages

The decanter centrifuge emerged as the optimal solution, offering several distinct advantages for whiskey distillery digestate processing. As illustrated in Figure 1, the system operates by subjecting digestate to high centrifugal forces typically ranging from 2000-4000 × g in a horizontally rotating bowl, achieving rapid separation based on density differences. The technology demonstrates excellent handling capacity for highly fluid feedstocks with total solids content of 1-5%, requiring no chemical conditioning while allowing adjustment of key parameters including bowl speed, differential speed, and weir height for process optimization.

Comparative studies demonstrate that decanter centrifuges achieve 15-25% higher solids capture efficiency than mechanical presses for food waste digestate while maintaining superior operational stability [6]. The technology's continuous operation capability and compatibility with automated control systems make it particularly suitable for industrial-scale applications where consistent performance and minimal operator intervention are critical requirements. The mechanical design incorporating wear-resistant materials in high-wear areas ensures durability when processing abrasive whiskey lees digestate.

4.3.2 Operational and Economic Benefits

The decanter centrifuge system provides significant advantages over alternative technologies across multiple operational parameters. As shown in Table 1, the technology offers an optimal balance between separation performance achieving 85-92% efficiency and operational costs with energy consumption of 1.5-3 kWh/m³. The absence of membrane fouling issues and reduced sensitivity to feed composition variations result in significantly

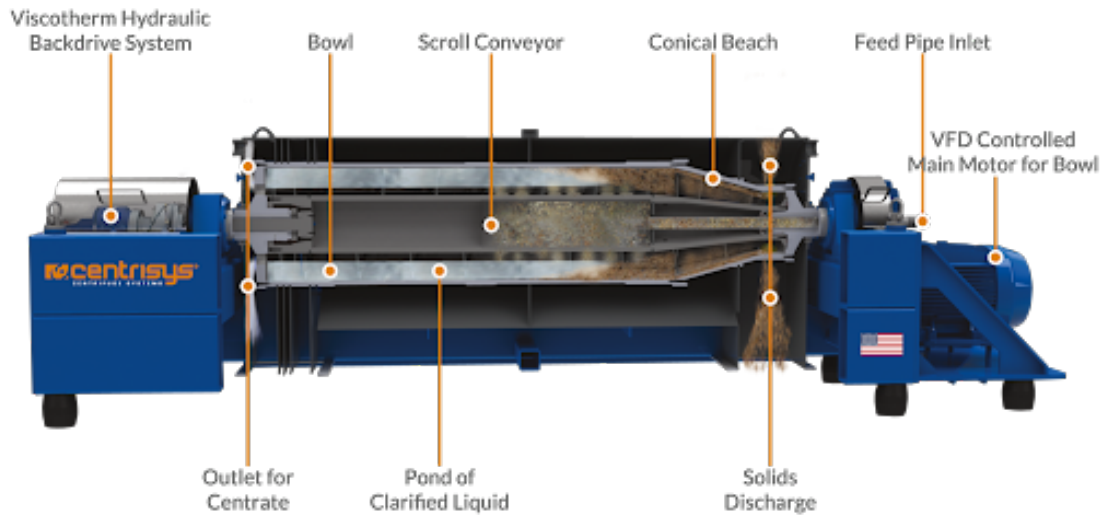


Figure 1: Schematic representation of decanter centrifuge operation showing principal components and material flows.

lower lifecycle costs, estimated at 30-40% reduction compared to RO systems [7]. Maintenance requirements are substantially reduced with monthly rather than daily interventions needed, while the compact footprint of 0.3-0.6 m² per m³/h capacity optimizes space utilization in the processing facility.

The output fractions are ideally suited for subsequent valorization pathways. The solid fraction with 18-25% total solids content demonstrates excellent characteristics for composting or biochar production, while the liquid fraction containing less than 1% total solids can be directly processed through ultrafiltration and nutrient recovery systems without risk of clogging or biofilm formation. This comprehensive compatibility with downstream processes enhances the overall system efficiency and resource recovery potential.

4.4 Alignment with Circular Economy Principles

The selection of decanter centrifugation technology strongly supports Biogas-ZE's sustainability objectives through multiple mechanisms. The system maximizes resource recovery by retaining 90-95% of nutrients in the output streams while completely eliminating the need for chemical flocculants or other additives. Energy efficiency improvements of 40-60% compared to membrane systems significantly reduce the carbon footprint of the separa-

tion process, while the technology's inherent compatibility with downstream valorization processes enables closed-loop material flows.

Recent studies by [2] and [6] confirm that mechanical separation via centrifugation represents the most sustainable approach for food waste digestate management, particularly when integrated with nutrient recovery systems. The technology's operational characteristics align perfectly with circular economy principles by enabling nearly complete recovery of both solid and liquid fractions in forms suitable for beneficial reuse. This approach transforms what would otherwise require disposal as waste into valuable inputs for agricultural and industrial processes, creating a model for sustainable distillery byproduct management in the beverage industry.

4.5 Conclusion

The implementation of decanter centrifugation technology provides an optimal solution for solid-liquid separation of whiskey distillery digestate, combining robust performance with variable feedstock quality and high separation efficiency without chemical additives. The system's low maintenance requirements make it particularly suitable for automated operation, while its excellent compatibility with downstream valorization processes ensures maximum resource recovery. These characteristics directly contribute to achieving Biogas-ZE's strategic objectives of zero waste processing, carbon

Table 1: Comparative analysis of separation technologies

Parameter	Membrane Filtration	Presses Mechanical	Decanter Centrifuge
Separation eff. (%)	90–95	60–75	85–92
Energy (kWh/m ³)	3–10	0.5–2	1.5–3
Chemicals	High	Moderate	None
Maintenance	Daily	Weekly	Monthly
Footprint (m ² /m ³ /h)	0.8–1.2	0.5–0.8	0.3–0.6

footprint reduction, and cost-effective resource recovery. The technology selection demonstrates how advanced mechanical separation can enable sustainable industrial-scale processing of organic byproducts while creating value from waste streams.

5 Liquid Fraction Processing

The liquid fraction of the digestate requires an iterative process when deciding on a fitting solution. The stages involved with this process include the initial filtration of the liquid fraction, the extraction of nutrients deemed desirable for valorizing, the resulting filtrate added to an algae photobioreactor meant to sequester carbon and produce biogas when added back to the anaerobic digester, and finally the final processing of the gray water after the algal biomass has been filtered out.

5.1 Biogas Production During Anaerobic Digestion of Algae

Through anaerobic digestion (AD), companies have been seeking a way to produce biogas while utilizing what is seen as waste in the process. The hope is a paradigm shift that allows for carbon sequestration and circularity of the food production chain. Currently micro and macro algae are the prime candidates given their incredible ability to fix carbon due to their quick growth time and large biomass. The anaerobic digested algae can produce biogas that contains ~50-70% methane, 30-45% CO₂, <2% H₂, <3.5% H₂S [8].

5.1.1 Macroalgae

The limiting factor for macroalgal rests in its special requirements which places a constraint on a facility's ability to cultivate and use it. Since macro algal represents a much higher biomass than microalgae, its capabilities to sequester carbon are far greater [9]. Microalgae also contains less lipids and carbohydrates which is beneficial if used as a substrate for biodiesel, however it is harder on the anaerobic digester environment. Although the high fixing rate makes macroalgae desirable, it has a slower growth rate.

5.1.2 Microalgae

Relative to macroalgae, microalgae contain a much higher amount of lipids and proteins which can lead to excessive ammonia production from the amino acid degradation [8]. However the environment can be adjusted through carbon additions to balance. For carbon sequestration, microalgae boasts up to 10 times more fixing than terrestrial plants and has a biomass productivity up to 50 times that of high yielding plants such a switchgrass [10].

The major difficulties for algal AD include the turbidity of the liquid fraction of the digestate re-entering the digester, the sulfur compounds algae contain (which will receive pretreatment), the nutrient balance, and the harvesting of the algal biomass.

5.2 Liquid Fraction Filtration Technologies

Multiple types of filtrations are currently used in wastewater filtration within the industry. One method of note includes ultrafiltration/nanofiltration using different membrane compositions such as ceramic hollow fiber and PET.

Centrifugal, reverse and forward osmosis systems are often found in the ranging literature used to filter wastewater. There is currently a sea of literature on wastewater treatment using these methods, however the pool of data that includes algae for anaerobic digestion is significantly smaller and the makeup of the wastewater differs from the feedstock in this design. Certain liberties are therefore necessary when assuming the validity of certain technologies and methods, such as heavy metal concentration, fouling and pathogen rejection, and filtering efficiencies. Keeping this in mind will help broaden the scope of potentials to be used in the design process.

To reduce the amount of filtration required overall, filtration of the liquid fraction as it comes out of the digester and before reaching the algal growth tank will be implemented. The initial thought was to utilize the nutrients to enhance the algae growth since they feed off of these nutrients, however it was deemed to be too complex in finding the balance of added nutrients and the subsequent extraction necessary of the surplus. This will alleviate additional filtration complications that would occur if nutrient extraction was set before this stage and will also limit the steps involved. Several considerations to be made are the overall filtrate turbidity which can prevent algal growth, macro-nutrient retention in the filtrate which is necessary for valorizing, and the operating life of the membranes used (as well as fouling). Another important challenge is if the membrane chosen captures the nutrients of interest, how will the membrane be treated in order to recover those important products.

5.2.1 Centrifugal Filtration

Frequently used in the harvesting of algae in the industry, centrifugal filtration raises a potential solution to filtering the biomass and particulate matter of the digestate. Considerations to be met are the species of algae used, the biomass percentage per batch and the high cost of power to run the operation. With an efficiency rating ~90 percent it is worth exploring as a method, however for this design the high cost and presence of better alternatives is hard to argue with [11]. Another issue with utilizing centrifugation is the resultant turbidity of the filtrate. The algae requires as

clear of a filtrate as possible since any particulates and turbid solution would restrict the access to the photobioreactor necessary for growth. With centrifugation comparatively left behind the other methodologies.

5.2.2 Reverse and Forward Osmosis

By using seawater (or a highly saline solution) as a drawing agent, forward osmosis represents an emerging wastewater filtration technology that is a low input (both energy and material) option to filter the liquid digestate. Through a naturally driven osmotic gradient, an impressive gamut of nutrients can be extracted with efficiencies surpassing 90 percent [12]. Unfortunately nitrogen and phosphorus are also removed through this method and thus make it difficult as being a useful tool to be implemented for this design [1]. In removing the nutrients meant for valorizing, the complexity of extraction would be increased and drive up costs. Other uncertainties existing such as membrane fouling and low specific usage data conclude the literature review of this method.

Unlike forward osmosis, reverse osmosis is a pressure driven method that may require certain pretreatments like primary filtering depending on the solution to be filtered. The resultant filtrate is excellent quality but fouling of the membrane persists as the largest hurdle in its usage as a filtration method. Once again this method also captures the nutrients in its membrane and requires secondary extraction/processing [13]. These two methods are great for potable filtered water as the end goal, however restrict the user to rely on a secondary round of treatment/processing to obtain valuable nutrients embedded in the membranes.

5.2.3 Ultrafiltration

Ultra- (or nano) filtration utilizes a two phase membrane that allows for almost a complete rejection of bacteria while retaining most of the nutrients present in the filtrate [14]. The ability to reject bacteria is of benefit specifically when waste streams are the primary feedstock as pathogens such as e.coli may prove problematic for both the digester and the photobioreactor. The result is a purified solution with nutrients available for either extraction or use for the algae growth.

Filter types span a diverse range of structure and material lending to a verifiable banquet of choices. Of interest are ceramic, recycled PET, and polymeric (PVC) based membranes in either hollow fiber structures or flat plate formations. The hollow fiber and flat plate structure does not seem to play a large role in the efficiency and ability to filter, so it is up to the designer to choose [15]. The main consideration on material type is potential for fouling and its life span. Especially since algae creates a high probability of fouling this parameter is of utmost importance. When comparing conventional membranes which are typically polymeric and relatively newer technologies such as ceramic it all comes down to this requirement. Both ceramic and PVC membranes reject bacteria and result in water with high purity, however ceramic ends up with much less fouling and can be cleaned for further use [16]. The cleaning of ceramic membranes however does entail a wastestream from whatever treatment is used, which will likely be chemical.

Recycled PET on the other hand is in its infancy but looks promising for future use as it has a high bacteria rejection rate, economic feasibility, and a large focus on promoting a circular economy through recycled plastics [17]. There is currently not enough data to support its use in scalable operations and no insinuations about the waste streams generated from its usage.

6 Nutrient Extraction: Struvite

As with most organic waste (digestate in this case), excessive solubilized nutrients is a standard that balances between beneficial and harmful. If untreated, digestate may be applied directly to nutrient starved fields safely and benefit the land it is treating. On the other hand, certain macronutrients can accumulate or leach leading to eutrophication of streamlines and waterways under and around the field of application. If the treatment is not diversified relative to its position this can harm an environment when the intention was good (or neutral). These nutrients have the ability to be processed into a marketable product that

supports a circular economy as well as benefiting the operators of such facilities.

Amongst the nutrients generally available in digestate, phosphorus and nitrogen compounds have the greatest future as a precipitated crystal formation (salt) known as struvite. Other nutrients exist of course but can actually hinder the growth/functionality of the anaerobic bacteria in the digester and the efficiency of precipitating struvite [18]. Alternatively these nutrients could be utilized as the growth support for the algae as the filtrate is added to the photobioreactor which was mentioned earlier in the filtration section. The major complication this presents is the ability to maintain the ratios of nutrients and the ending recovery of surplus after the growth tank.

6.1 MAPs (Struvite)

Seemingly diamonds in the rough, struvite is the ultimate byproduct of anaerobic digestion (AD) that creates an economically feasible circular waste management method. Unfortunately it is not that simple as it is one of the largest complicators of AD functionality due to precipitation of struvite during the process. Its chemical composition of magnesium, ammonium, and phosphate ($\text{MgNH}_4\text{PO}_4 \cdot 6\text{H}_2\text{O}$) in equal molar amounts, and when the concentrations raise above its solubility in alkaline environments (starting around pH 8, such as in some AD) it will spontaneously precipitate. This spontaneous precipitation is undesirable and so pH control is the best route to take with some papers referencing pHs of 9+ as the optimal precipitate conditions [19]. This means keeping AD under this pH is paramount to maintaining solubilized forms of the salt and restricted loss of valuable products.

6.2 Extraction Methods and Technology

The precipitation of struvite occurs when the ratio of MAP present in the digestate rises above its stoichiometric ratio [19]. This requires several different components for success such as low competition, reagents, and a reactor for the process. Generally feedstock for anaerobic digesters contain varying levels of magnesium, potassium, and cal-

cium which all compete to form crystals and can inhibit or lower the struvite formation that is of value [18]. This entails that some processing may be necessary to remove the competitors out if they are present in large concentrations.

To extract the struvite most operations utilize a magnesium compound (frequently magnesium oxide, MgO) since it is generally the limiting factor for the stoichiometric ratio. Additional supplementation of either nitrogen or phosphorus may be needed as well depending on the concentrations of the digestate. Other reagents have been used such as flocculants, however magnesium remains an industry standard, mostly varying only in the Mg compounds chosen as the reagent. Flocculants create a large waste stream and also present complexities for further separation of the struvite. Alterations to pH are also necessary and will require a solution to adjust such as sodium hydroxide (NaOH). Struvite nucleation was improved by seeding before initial formation was undertaken [20].

6.3 Reactor Types for Struvite Formation

As mentioned, a reactor for the process to take place is needed. There are several options like stirred tank, fluidized bed, and biomembrane based reactors with different morphologies, pros and cons. Bio based reactors primarily are used to extract phosphorus without precipitating struvite and display difficulties in scaling up for non-lab based operations and high fouling rates of the membranes, therefore they will not be investigated further [21]. This choice was also defended in the reasoning for filtration methods due to the complexity of extraction from the membranes. Further considerations to take into account are whether the reactor is operated continuously or in batches, as well as the efficiencies of over struvite extraction. The two most commonly used reactors are stirred tank (completely mixed) and fluidized beds and their functionality varies based on the desired mode.

6.3.1 Stirred Tank Reactors

Stirred tanks, as seen in [22, Fig. 2] can operate as either a batch or a continuous process by separating the struvite formation and precipitation phases

[19]. The efficiency and size of the crystals ends up becoming a function of the stirring speed (increased efficiency in the turbulent zones created) [22]. Some papers cited extraction percentages for ammonium, magnesium and phosphate around 92%, 75%, 91% respectively, whereas for fluidized bed reactors at 89%, 90%, 76% with a much lower water content [23]. The ease of control for the mixing speed, influent/effluent discharge rate, and reactants mixing represent a huge advantage and consistent product [24]. The substantial disadvantages lie with the large energy costs due to the power needed for high stirring speeds, and the lack of crystal dimension control.

6.3.2 Fluidized Bed Reactors

Operating in a continuous flow mode, fluidized bed reactors allow for optimal mixing of the influent and have similar flexibility and control to STR. In [25, Fig. 3] a general schematic of a fluidized bed reactor is shown. The greatest benefit of using FBR is the low energy consumption relative to STR, as well as its smaller environmental impact and potential for pelletized products [23]. Mitigation required for FBR are the fine crystals formed as they can remain in the effluent without being properly extracted and the control of pH around 9 [25].

6.4 Final Processing

Once the filtered, nutrient-extracted liquid fraction reaches the algal tank there is the potential to be amended if the nutrient load is required for algal growth. This is assuming there are no added nutrients into the photobioreactor other than the digestate. When exiting the algal tank it will need to be filtered once again to remove the algae and particulate matter. This will result in two waste streams, gray water and wetted solid biomass. This gray water can be either reprocessed into potable water for recycling back into the system or used to adjust the water content of the feedstock entering the digester. Other uses for the gray water will change the type of filtration needed since regulations of wastewater differ regionally and the end goal may remain as disposal. The wet solid matter will be sent for pretreatment (to ensure sulfur compounds and biomass structure

will degrade) and can be reused within the digester as feedstock.

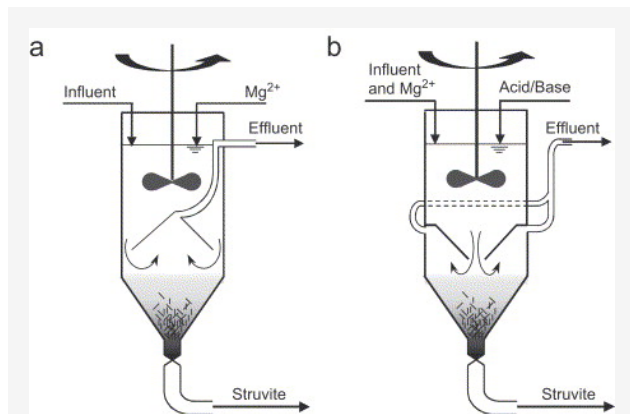


Figure 2: Stirred tank reactor schematic showing the pathways for influent, effluent, struvite and reagents

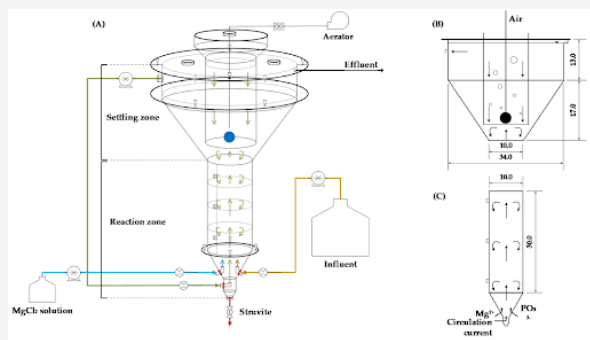


Figure 3: Fluidized bed reactor schematic showing the pathways for influent, effluent, struvite and reagents

7 Solid Fraction Considerations

Digestate is the byproduct of anaerobic digestion, a promising waste-to-energy technology that converts organic material into biogas while producing a nutrient-rich residue [26]. The physical separation of digestate typically produces a liquid fraction and a solid fraction, with the latter containing between 20-40% dry matter depending on the separation technology used [27]. The solid fraction of digestate presents unique challenges and opportunities in the context of waste management and resource recovery. Here, we will explore different processes considered in the project to determine the best use for the solids' fraction of the digestate.

A decision matrix was compiled to help visualize all the different factors affecting the resulting decisions made by the team, this can be seen in Fig. 7.

7.1 Characteristics of Digestate Solids

Mechanical separation through decanter centrifuge is a common method to separate digestate into solid and liquid fractions. While this process produces a relatively clear liquid fraction, the solid fraction still retains considerable water content, typically around 60% [28]. This high moisture content presents significant challenges for transportation, storage, and further processing [27].

The solid fraction is characterized by elevated levels of organic matter and phosphorus, organic nitrogen compounds, fibrous and recalcitrant organic materials, along with variable quantities of heavy metals and potential pathogens. This untreated material remains biologically active, which may result in progressive degradation, odor problems, and pathogen-related risks [29].

Proper management of the characteristics of digestate solids must address multiple regulatory frameworks governing their handling, processing, and end use. These regulations typically establish limits for pathogen concentrations, heavy metal content, antibiotic or pharmaceutical residues, and application rates to prevent nutrient leaching and runoff. Nutrient leaching and runoff from improperly managed digestate solids can contribute to eutrophication of water bodies, representing a significant environmental concern [30]. Additionally, facilities must consider emissions of ammonia, greenhouse gases, and odorous compounds during processing and storage [31].

7.2 Key Objectives in Processing Digestate Solids

Effective processing of digestate solids generally aims to achieve several objectives simultaneously. Stabilization is essential for reducing biological activity to prevent further degradation and associated emissions. Pathogen reduction is necessary for meeting regulatory requirements for safe handling

and application. Valorization converts digestate into marketable products, while volume reduction minimizes transportation and storage costs. Emission control is crucial for limiting environmental impacts during processing, and energy efficiency optimizes resource utilization throughout the treatment chain.

7.3 Key Selection Criteria for Processing Methods

7.3.1 Environmental Impact

When evaluating processing methods, environmental performance represents a critical consideration, particularly for facilities pursuing zero-emission goals within a circular economy framework [32]. Key environmental factors include emissions profile, which covers direct and indirect emissions of greenhouse gases, ammonia, and odorous compounds; waste stream generation, which involves volume and characteristics of secondary waste requiring further management; and water recovery potential, which presents opportunities for water recycling to reduce freshwater consumption, particularly important in the region of this facility, which has hard water conditions that require costly treatment.

Direct disposal of process water into local waterways typically requires extensive treatment to meet discharge regulations, while municipal wastewater facilities may have limited capacity to accept high-volume industrial effluents from this facility [28]. Therefore, processing methods that minimize wastewater generation or enable water recovery present distinct environmental advantages.

7.3.2 Energy and Efficiency Requirements

The operational efficiency of processing methods significantly impacts the sustainability and economic viability of digestate treatment [33]. Residence time is an important factor, as shorter residence times facilitate continuous processing, improve scalability, and reduce facility footprint requirements. Long residence time processes can present significant space constraints and operational complexities for large-volume applications [28]. Energy input is another consideration, as

processing methods vary considerably in their energy demands, affecting both operational costs and environmental footprint. Integration with existing energy systems, particularly utilization of waste heat from industrial processes already occurring on the site, can dramatically improve overall efficiency [27]. Scalability and modularity provide the ability to scale operations, offering operational flexibility and reducing initial capital risks, particularly for pilot projects with expansion potential [32].

7.3.3 Economic Viability

Economic considerations ultimately determine the feasibility of implementation, in which several factors need to be considered [33]. Capital expenditure involves initial investment requirements, including equipment, infrastructure, and installation costs. Operational expenditure covers ongoing costs for energy, labour, consumables, and maintenance. End product marketability reflects revenue potential from produced materials, with established markets presenting lower commercial risk than novel products. Regulatory compliance costs represent expenses associated with meeting applicable regulations, including testing, reporting, and potential permit requirements.

Processes that generate standardized products with existing market demand typically offer more predictable revenue streams than those producing novel outputs requiring market development [34].

8 Solids Processing Methods

8.1 Low-Processing Methods

8.1.1 Direct Land Application

Direct land application represents the simplest approach to digestate solids management, involving minimal processing before spreading the digestate onto land for agricultural use [30].

While direct application maximizes nutrient recycling with minimal energy input, it presents significant environmental risks and often faces regulatory barriers [30]. The high moisture content of untreated digestate creates transportation inefficiencies and application challenges. Seasonal

restrictions and usage on land application further complicate this approach, potentially requiring substantial storage capacity [29].

8.1.2 Composting

Composting involves the aerobic decomposition of organic matter under controlled conditions, producing a stable humus-like product, high in nutrients beneficial for plants and soil health [35]. This process utilizes microbial activity to degrade, stabilize, and deodorize organic substrates through mesophilic and thermophilic microorganisms.

Composting offers several advantages, including production of a marketable end product with established applications in agriculture, landscaping, and home gardening [35]. However, the process is relatively slow (typically 30-180 days), space-intensive, and requires careful management to maintain optimal conditions and prevent odour issues [29].

Biodrying and hyper-thermophilic composting (HTC) have shown particular promise for accelerating digestate stabilization while reducing greenhouse gas emissions during processing. Research indicates these approaches can achieve 27-84% nitrogen fixation when combined with appropriate chemical or biological treatments [4].

8.2 Thermal Conversion Methods

8.2.1 Direct Combustion

Direct combustion involves burning dried digestate to generate energy, resulting in mineral ash residue [36].

Studies have demonstrated that digestate can be co-combusted with wood in a 1:1 ratio for use in conventional furnaces after appropriate drying and pelleting [36]. While combustion achieves complete volume reduction and energy recovery, it eliminates the soil amendment value of digestate and requires significant energy input for pre-drying [37].

8.2.2 Pyrolysis

Pyrolysis thermally decomposes organic material in an oxygen-limited environment at temperatures typically between 400-800°C, producing biochar, syngas, and bio-oil [37].

Biochar is a carbon-rich material produced through the pyrolysis of biomass under oxygen-limited conditions, serving as a stable soil amendment that enhances fertility and sequesters carbon [38]. Its production results in a porous structure that improves water retention and nutrient availability in soils [39]. Additionally, biochar contributes to climate change mitigation by locking carbon in a recalcitrant form, reducing greenhouse gas emissions from decomposing biomass [40]. However, significant environmental concerns exist regarding the aqueous solution generated during pyrolysis, which contains phenols, furan derivatives, and other potentially toxic compounds requiring treatment [37].

Syngas, or synthesis gas, is a mixture of hydrogen (H_2), carbon monoxide (CO), and other gases generated via gasification or partial combustion of carbonaceous materials, including biomass, coal, or waste [41]. This versatile energy carrier can be utilized for electricity generation, hydrogen production, or as a feedstock for synthetic fuels and chemicals [42].

Bio-oil is a dark, viscous liquid obtained from the fast pyrolysis of biomass, where organic materials are rapidly heated to 400–600°C in the absence of oxygen, followed by quick condensation of vapors [43]. While bio-oil has potential as a renewable fuel, its high oxygen content and acidity necessitate upgrading through hydro-deoxygenation or catalytic cracking to improve its stability and energy density [44]. Alternatively, bio-oil can be refined into chemicals or used directly in boilers and turbines, though challenges related to storage, corrosion, and combustion efficiency remain [45].

8.2.3 Hydrothermal Carbonization (HTC)

Hydrothermal carbonization processes wet biomass under high pressure (1000-3000 kPa) and moderate temperatures (180-250°C) to produce hydrochar through subcritical water reactions [46].

HTC offers the advantage of processing wet digestate without pre-drying requirements [46]. However, the process generates acidic hydrochar containing polycyclic aromatic hydrocarbons (PAH's) and other organic toxic compounds that may limit application potential [47]. Additionally, the liquid

fraction produced during HTC contains various organic acids and phenolic compounds requiring further treatment [37].

8.3 Drying and Concentration Methods

8.3.1 Conventional Thermal Drying

Conventional drying applies direct or indirect heat to digestate solids, typically using rotary drums, belt dryers, or fluid bed systems [48]. Thermal drying provides rapid processing and significant volume reduction, creating a stable, easily transportable product [48]. However, the process demands substantial energy input, approximately 13 kWh of electricity per cubic meter of digestate processed [28]. Integration with waste heat from biogas engines can improve efficiency, with approximately 1.4 litres of water removed per kilowatt-hour of thermal energy [48]. However, this process still remains energy intensive, and not ideal for zero emissions or sustainable process.

8.3.2 Biodrying

Biodrying utilizes heat generated by aerobic microorganisms combined with controlled aeration to reduce moisture content in organic waste [49]. This process represents a distinct approach from composting, focusing on drying and preserving biomass content rather than complete stabilization [49]. Commercial biodrying cycles typically complete within 7–15 days, achieving mass losses of approximately 25–30% w/w primarily through water evaporation and CO₂ release [49].

Beyond moisture reduction, biodrying can retain nutrients in organic substances and reduce hazardous pathogens through thermophilic microbial activity [4]. The process requires careful management of airflow rates and, in some systems, dehumidification of recirculated process air to enhance drying efficiency [49].

8.3.3 Vacuum Evaporation

Vacuum evaporation involves boiling digestate under reduced pressure (20–50 kPa), allowing moisture removal at significantly lower temperatures (60–80°C) than conventional drying [5, 28]. The reduced temperature requirement enables efficient

utilization of waste heat. This thermal integration substantially decreases the overall energy demand for the evaporation process. [5, Fig. 4] illustrates the fundamental working principle of the vacuum evaporation system.

The liquid digestate first enters the evaporation chamber where the applied vacuum reduces the boiling point to approximately 60°C. As the temperature rises, water evaporates while dissolved gases such as carbon dioxide and ammonia are released from the liquid phase. Due to the high ammonium content typical in digestate, the resulting steam contains significant amounts of ammonia that must be removed to produce clean water.

This ammonia removal is accomplished through an acid scrubbing system, where sulfuric acid reacts with the ammonia to form ammonium sulfate. Following this treatment, the purified steam passes through a condensation unit where it is converted back to liquid water and stored in a holding tank. To complete the thermal cycle, the stored water circulates through an evaporative cooler to dissipate residual heat from the system.

The energy requirements for this process are approximately 13 kWh of electrical energy per cubic meter of processed digestate, while the thermal energy efficiency reaches 1.4 litre of water removed per kilowatt-hour of thermal energy input [5].

For enhanced energy efficiency, multiple evaporation stages can be implemented. In such configurations, different pressure levels are maintained in successive evaporators to create a temperature gradient. The steam generated in the highest-temperature evaporator transfers its thermal energy to heat the digestate in the subsequent evaporator operating at lower temperature and pressure. Commercial systems typically employ up to three evaporation stages, creating a cascading thermal recovery system that significantly improves overall energy efficiency compared to single-stage operation [5].

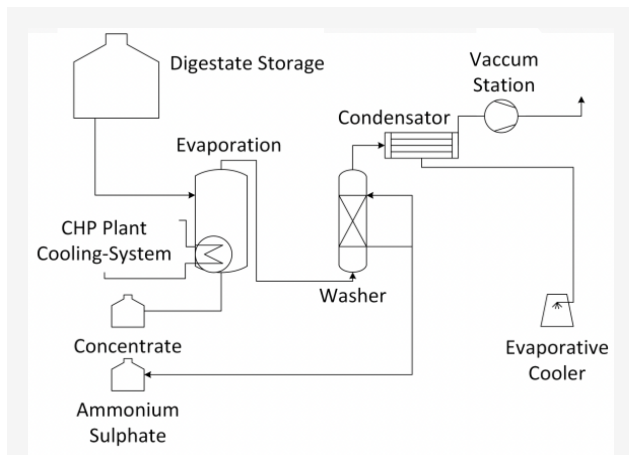


Figure 4: Overview of a generic vacuum evaporation system

9 Recommended Method: Vacuum Evaporation

Based on comprehensive analysis of available processing methods, vacuum evaporation emerges as particularly promising for facilities with available waste heat and water recovery objectives [28, 50].

9.1 Key Trade-offs in Method Selection

The selection of appropriate processing methods involves navigating several fundamental trade-offs. Energy intensity versus processing speed represents one trade-off, as high-energy thermal processes typically achieve rapid results, while biological methods require less energy but longer residence times [27]. Capital versus operational expenses is another consideration, as technologies with lower capital costs often entail higher operating expenses, while more complex systems typically require greater initial investment but may reduce long-term operational costs [33]. Product marketability versus process complexity presents another trade-off, as established products like compost have predictable markets but limited value, while more sophisticated processes may yield higher-value products with less certain market demand [34]. Environmental performance versus economic efficiency is also important, as processes with superior environmental profiles may entail economic trade-offs that must be evaluated within the broader context of

facility sustainability goals [32].

9.2 Advantages Over Alternative Methods

Vacuum evaporation offers several distinct advantages over alternative approaches. Compared to composting, it offers significantly faster processing time and smaller footprint, critical factors for industrial-scale operations with continuous processing requirements [50]. Compared to thermal conversion methods, it provides lower emissions profile, superior nutrient preservation, and reduced complexity without secondary waste streams requiring treatment [28]. Compared to conventional drying, it enables more efficient energy utilization, particularly when integrated with waste heat recovery systems, and superior nitrogen retention [50, 51]. A full decision matrix is presented in Fig. 7.

9.3 Alignment with Key Selection Criteria

Vacuum evaporation demonstrates strong performance across the established selection criteria. Its environmental performance includes water recovery potential, near-zero emissions when properly designed, and closed-loop capability that aligns with circular economy principles [50]. For energy efficiency, when integrated with waste heat utilization, the system provides sufficient capacity to process the entire digestate output, unlike many alternative technologies [5, 50]. It also offers operational flexibility, as modern vacuum evaporation systems offer modular designs that facilitate scaled implementation and operational adaptation [28]. The product quality achieved with vacuum evaporation results in concentrated product with approximately 10-25% moisture content, suitable for various end applications while retaining up to 97.5% of nitrogen value [50, 51].

9.4 Implementation Considerations

Successful implementation of vacuum evaporation systems requires attention to several critical factors. Heat integration represents a critical consideration,

as proper integration with existing heat sources dramatically impacts economic viability [5, 28]. pH control is also vital, as maintaining acidic conditions ($\text{pH} \approx 5.0$) during evaporation significantly improves nitrogen retention [50, 51]. System sizing needs careful attention to ensure appropriate capacity matching with digestate production volumes prevents operational bottlenecks. Materials selection should prioritize corrosion-resistant materials required for long-term reliability with acidified digestate [28]. Condensate management is necessary to establish appropriate applications for recovered water to maximize system benefits [50, 51].

Processing the solid fraction of digestate represents a critical challenge in developing sustainable anaerobic digestion systems. While multiple technologies offer viable approaches, vacuum evaporation emerges as particularly promising for facilities with waste heat availability and water recovery objectives.

The method effectively addresses the fundamental challenges of digestate management: stabilization, volume reduction, pathogen control, and nutrient preservation. Its ability to recover water while concentrating nutrients aligns with circular economy principles, while modular implementation supports scalable development strategies.

Further research opportunities exist in optimizing multi-stage evaporation systems, developing specialized applications for condensate streams, and enhancing the marketability of concentrated digestate products. As regulatory frameworks evolve and markets develop, integrated approaches combining complementary technologies may offer increasingly sophisticated solutions to digestate management.

10 Disposal of the Dried Solid Product

Following vacuum evaporation, the primary options considered for the disposal of the dried solids were combustion and land application. In the literature review process, several other options for management were available, including pyrolysis and composting. These methods were introduced and

eliminated in sections 8.1.2 and 8.2.2, respectively. Notably, the primary options considered were also included in Section 8, however were re-evaluated following the drying of the product via vacuum evaporation. This processing step addressed some of the initial limiting factors to the implementation of direct application and combustion, namely the high moisture content causing storage and transportation difficulties with direct field application. Additionally, combustion is considered in detail in Section 8 as the drying step is necessary prior to its implementation. Ultimately, we propose land application as the recommended method for disposing of the concentrated solid product, however a review of combustion is also included.

10.1 Key Objectives in Disposing of Dried Digestate Solids

The effective disposal of the dried solid product consists of a few key considerations. Environmental compatibility ensures products are environmentally benign or contribute positively to sustainability goals. Regulatory compliance focuses on meeting environmental standards and legal requirements throughout the disposal process. Emission control aims at managing waste to limit emissions related to transportation and treatment. Cost minimization works toward reducing disposal expenses through value-added processing or achieve net-zero cost outcomes. These considerations were used as a basis for selection of the method of disposal of the dried solid product.

10.2 Combustion

The combustion of the dried solid product as a biofuel represents a potential valorization pathway, offering both energy recovery and waste reduction benefits; however, the high ash content of digestate poses operational challenges that may require specialized equipment, resulting in significant capital expenditure.

A study on the combustion behavior of digestate produced from apple pomace and maize silage [52] found that digestate derived from these agricultural residues can serve as a viable biofuel, with

calorific values between 15.34 MJ/kg and 18.41 MJ/kg depending on the feedstock blend. However, the digestate's ash content in this study ranged from 8.2% to 11.6%. These calorific values are roughly on par with reported calorific values of woody biomass pellets currently on the market, however the ash content is significantly higher. ISO 17225-2 standards requirements for graded wood pellets used for industrial and non industrial use specify that the ash content should not exceed 2% [53].

The high ash content is a major limiting factor for the combustion of digestate as a biofuel, as it could lead to operational challenges such as ash sintering and slugging in combustion systems which could cause combustion system failures if specialized equipment is not used. This represents a high capital expenditure for the distillery, and necessitates a legislative review to determine proper emissions limits and potential concerns of air pollution based on the specific properties of the dried solids.

The ash produced during combustion could be rich in nutrients including potassium, calcium, magnesium, sodium, and phosphorus [52]. This composition supports the potential for ash reuse as a fertilizer or soil amendment, offering a secondary benefit to the combustion process. However, material specific analyses would need to be done to determine whether the ash could be sold as a valorized product, and to determine the ash content of the dried solids. If combustion is to be used to manage the dried solid product, a full economic analysis would need to be done on the tradeoffs between energy production through combustion and the capital expenditures that would be necessary to produce the energy, as well as considerations for the value of the final product.

10.3 Land Application

The recommended solution for the disposal of the dried solids in this case is land application as an organic amendment due to its high organic matter content and biological stability. The precise composition of dry digestate solids is dependent on numerous factors, such as the AD conditions, the storage style, the extent of post digestion re-

finement, and the characteristics of the initial feedstock [54, 55]. Additionally, the effect of the material on the soil is the type of digestate and its chemical composition, the layers of the soil profile and the individual properties of each layer [55]. These factors are site and situation specific, necessitating that both site specific soil analyses be done and that full analyses of the dried solid characteristics are conducted. Despite uncertainties around the exact nutrient composition of the material, the addition of the dried solids as a soil amendment has been demonstrated in literature as being beneficial for increasing soil organic matter, contributing to soil aeration, and contributing to stores of nutrients in the soils. Land application of the dried solids aligns with the ethos of Biogas-ZE of recycling material streams back into production, in this case the production of agricultural goods, and minimizing waste streams.

10.3.1 Nutrient Management

Although the separation of digestate typically extracts most nitrogen into the liquid phase [56], the remaining solids can still provide adequate nitrogen to promote microbial activity, particularly through the use of vacuum evaporation as a drying method which has been linked to strong nitrogen preservation [50, 51]. Soil treated with solid digestate has shown increased amounts of nitrogen-containing microbial biomass, suggesting that the pelletization and field application of the solid digestate can contribute to nitrogen immobilization in the soil [55]. This microbial activity from the addition of the dried solids has beneficial effects on yield in the long term in nitrogen limited soil due to the stimulation of biotic processes and increase of SOM [57]. However, depending on the specific composition of the material, the immediate nutrient demands of the crops may not be met by the addition of the dried solids alone. In such cases, supplemental nitrogen from sources like manure or mineral fertilizers may still be necessary [58]. Ultimately, the impact of the application of the dried solids on soil carbon sequestration and nutrient runoff depends strongly on both the characteristics of the soil and the material being applied [59].

The application of digestate as an amendment to agricultural land must be managed through a nutri-

ent management plan (NMP) to prevent nutrient losses and protect nearby ecosystems [60]. In Kentucky, any farm applying fertilizers, manures, or other soil amendments on more than 10 acres is required to implement an NMP as part of the Kentucky Agriculture Water Quality Plan [61]. These plans are meant to ensure long-term sustainability by preventing nutrient pollution. The structure recommended by [60] and [61] represents a set of best management practices for nutrient management at a farm scale that would need to be carried out for all farms receiving the dried solid product. These practices include site assessment gathering field-specific information on acreage, slope, soil type, and minimum distances from water sources, as well as results from soil analyses; digestate assessment providing information about digestate volumes, nutrient concentrations, storage infrastructure, and AD feedstock; application of the 4-R framework focusing on right source, right place, right amount, and right time; implementation of land management strategies to limit erosion and runoff; and establishment of recordkeeping and reporting systems to document nutrient applications, crop yields, weather conditions, and any changes to the plan. It is likely that many of the farms already have NMPs in practice and would simply need to adjust them accordingly.

10.3.2 Addressing Risks

Although the land application of the dried solids provides agricultural benefits, there are risks associated with its use that must be addressed, including soil acidification, heavy metal content, pathogens, pharmaceuticals, hormones, personal care products, and microplastics. Many of these contaminants are generally associated with digestate produced from sewage, municipal waste, industrial waste, or animal waste feedstocks, meaning they present less of a threat in this case where the feedstock for AD is stillage [62]. While pathogens and soil acidification may still pose risks, most pathogens and viruses would be inactivated inside the digester, and soil acidification can be addressed by liming, which is a common practice in agriculture. To ensure safety, the presence of contaminants in the dried solids will need to be monitored to ensure that the presence of any unfavorable

compounds is below regulatory limits set by the State of Kentucky.

10.3.3 Storage and Transportation

Effective storage of digestate is critical to facilitate its use as a fertilizer while minimizing environmental losses. To align nutrient availability with crop uptake, digestate must be stored until periods of optimal field application. Some crops require nutrient application at specific times, for instance at seeding for corn and row crops, while some crops have a more flexible window for application, such as perennial crops like pasture or hay. On many farms in Kentucky, there are fields where organic amendments can be applied throughout most of the year, with the exception of winter when frozen ground conditions prevent spreading [61]. Consequently, the design and location of storage infrastructure must account for both seasonal timing and the volumes generated.

A study assessing storage strategies for a distillery in Ireland compared two primary approaches: centralized and decentralized storage systems [63]. In a centralized model, all digestate is held in a large, single facility at the biogas plant. The on-site storage tanks for the dried solid product would need to be large to accommodate the full volume, as many of the truck movements for the distribution of the material would occur within a limited period to align with optimal agricultural application periods. Conversely, the decentralized model utilizes distributed storage tanks throughout the surrounding area, with individual tanks located in regions where digestate will be applied. This approach involves ongoing transport of digestate from the AD facility to the decentralized tanks throughout the year, enabling more flexible, localized application schedules.

We recommend decentralized storage be used to hold the dried solid product until it can be field applied. While both options result in similar annual truck movements, decentralized storage reduces traffic congestion by spreading transport over time, rather than concentrating it within the brief application windows. Additionally, decentralized storage also offers modular expansion of the system's capacity. If digestate production in-

creases due to expanded operations, as anticipated in the memorandum of understanding between BiogasZE and the Log Still Distillery, storage can be incrementally added across the network without the need for major upgrades to a centralized site. Specific responsibilities for tank sizing and storage could be managed by the distillery, participating farmers, or a third party depending on operational arrangements.

11 Conclusion

The digestate management system designed for Biogas-ZE delivers a comprehensive solution that optimizes environmental performance, operational efficiency, and resource recovery. The final processes which are recommended in this report can be seen in Fig. 6. Through the integration of decanter centrifugation for solid-liquid separation, ultrafiltration and struvite crystallization for liquid fraction processing, and vacuum evaporation for solid fraction management, the system achieves multiple sustainability objectives simultaneously. The selected technologies demonstrate superior performance across critical parameters including energy efficiency, nutrient preservation, operational reliability, and product quality. For solid fraction management, vacuum evaporation emerged as the optimal solution based on systematic evaluation against established criteria. The technology enables efficient moisture reduction while preserving nutrient content, particularly nitrogen which is retained at rates up to 97.5%. The system's integration with waste heat recovery mechanisms significantly improves energy efficiency, supporting Biogas-ZE's emission reduction objectives. Land application of the dried solid product is recommended as the most sustainable disposal pathway, creating a circular nutrient flow while providing agricultural benefits through soil organic matter enhancement and nutrient contribution. The liquid fraction processing incorporates ultrafiltration for initial clarification followed by parallel valorization pathways. Struvite precipitation recovers valuable phosphorus in a crystallized form suitable for commercial fertilizer applications, while algae cultivation enables carbon sequestration and pro-

duction of biomass suitable for energy recovery through anaerobic digestion. This multi-pathway approach maximizes resource recovery while creating closed material loops within the system. This project demonstrates that digestate from whiskey distillery waste represents a valuable resource when managed through appropriate technological interventions. By implementing systems that transform waste streams into agricultural inputs, Biogas-ZE can achieve its sustainability objectives while contributing to regional agricultural productivity and resource conservation. The framework developed herein establishes a model for similar facilities seeking to enhance the environmental and economic performance of anaerobic digestion processes within a circular economy context.

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A Figures

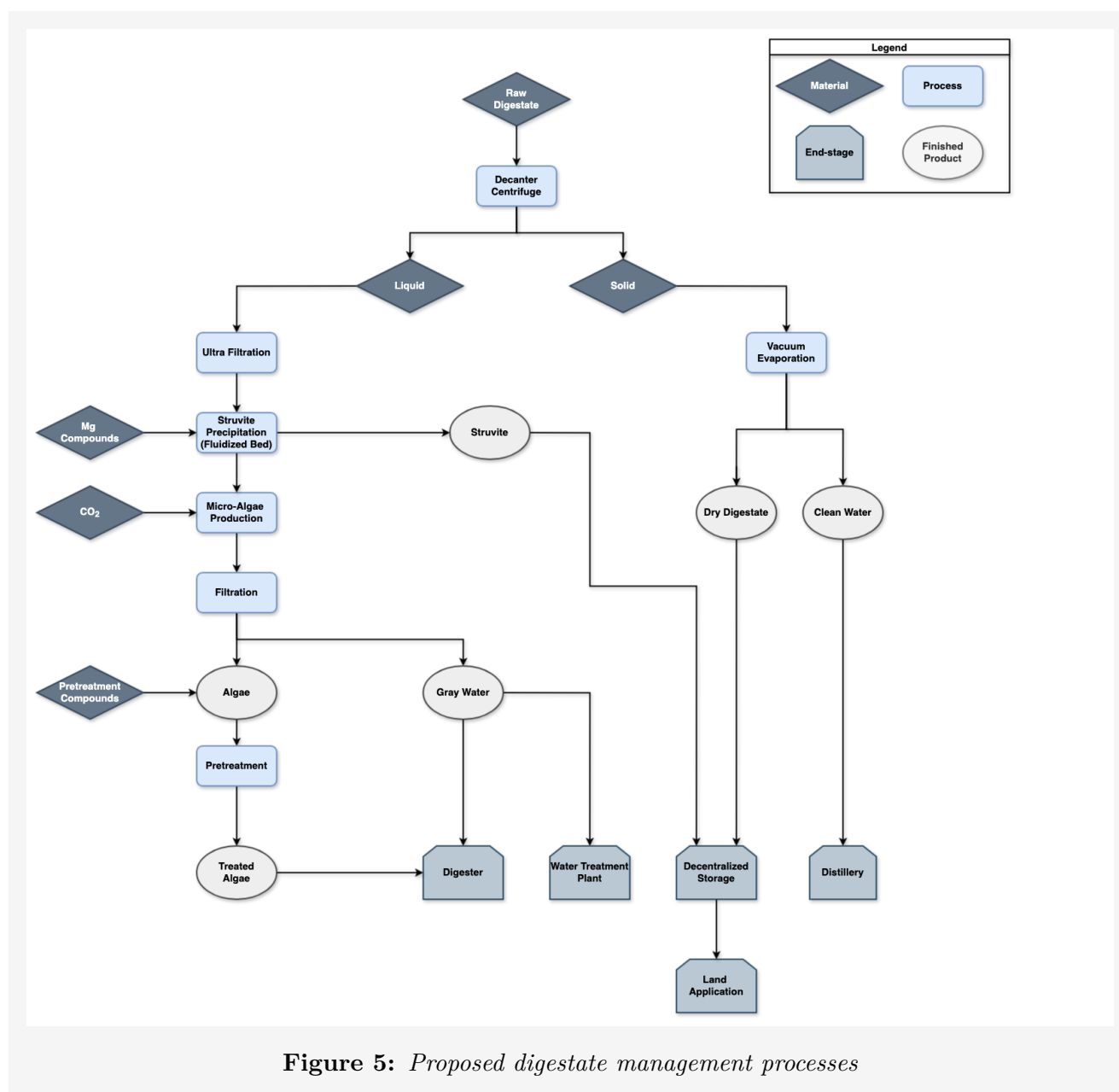


Figure 5: Proposed digestate management processes

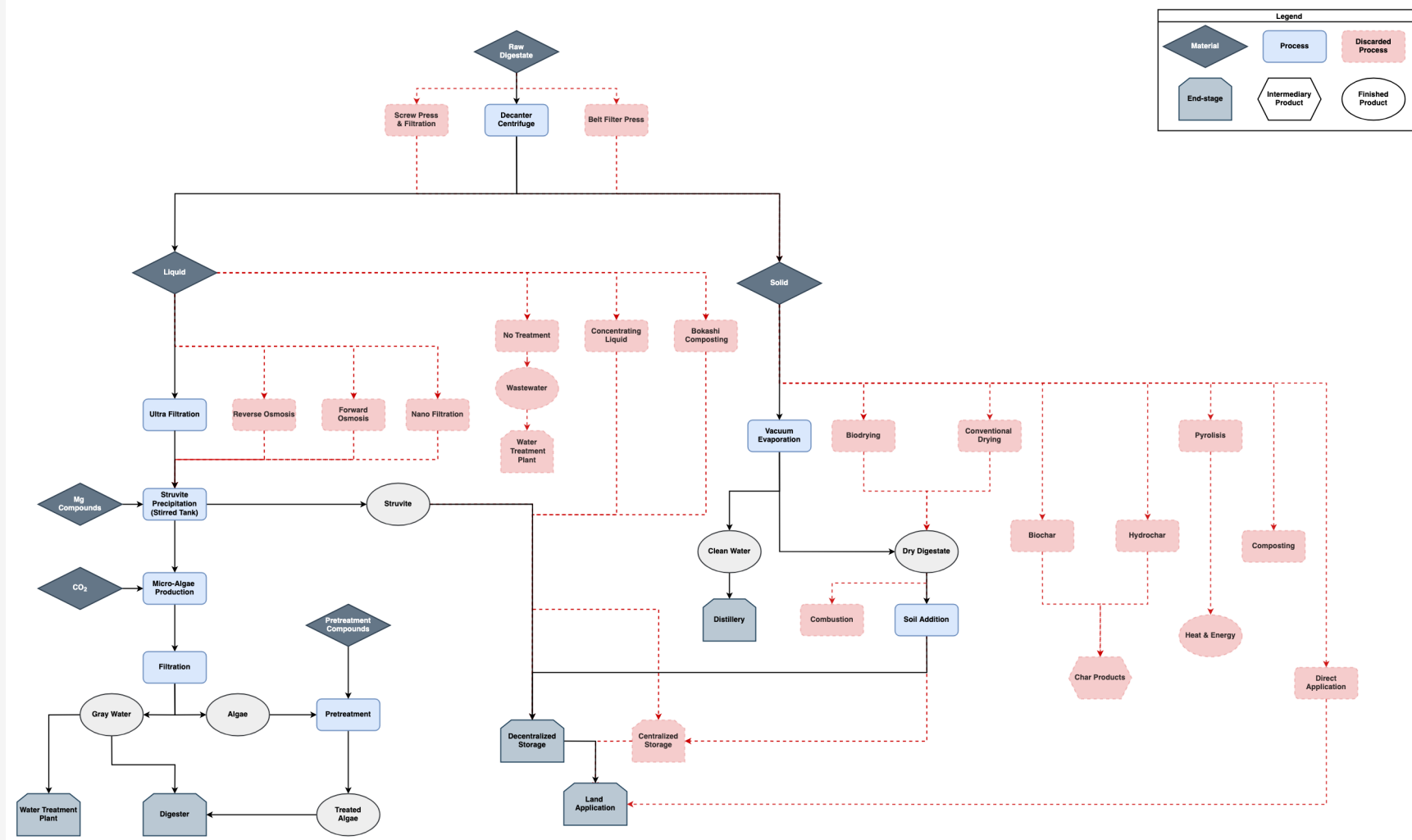


Figure 6: Process decision flowchart containing all processes considered as part of this project

Process	Technical				Environmental			Economic		Operational				By products
	Energy Input	Waste Stream	Residency Time	Scalability	Pathogen Reduction	Emissions	Water Recovery	Market Value	Capital Cost	Product	Complexity	Regulatory	Nutrient Retention	Byproducts
Direct Land Application	Low	No	Immediate	High	Low	Medium	No	Low	Very Low	N/A	Very Low	High	Medium	No significant byproducts
Composting	Low	Potentially	Months	Medium	High	Medium	No	Medium	Medium	Compost	Medium	Moderate	High	CO ₂ , methane (if uncontrolled)
Pyrolysis	High	Yes	Hours–Days	High	High	High	No	Medium	High	Biochar	High	Complex	Med–Low	Volatile gases (usable as fuel)
HTC	High	Yes	Hours	High	High	Medium	No	Medium	Very High	Hydrochar	High	Emerging	Medium	Organic acids, CO ₂
Conventional Drying	Very High	Potentially	Hours–Days	High	High	Medium	No	Medium	High	Dried Pellets	Medium	Established	Low	Water vapor, potential organics loss
Biodrying	Low	No	Days–Weeks	Medium	High	Low	No	High	Medium	Soil Amend. Dried Pellets	Medium	Low	High	CO ₂ , odor emissions (controlled)
Vacuum Evaporation	Medium	No	Hours–Days	High	High	None	Yes	High	High	Soil Amend. Dried Pellets	Med–High	Low	High	Concentrated liquid (usable or treatable)

Blue: Favorable/Positive aspect Yellow: Moderate/Neutral aspect Orange: Unfavorable/Negative aspect

Notes: Colour coding compares relative performance within each column. Market value depends on regional demand. Emissions assume proper system controls. Byproducts may vary based on process parameters and system controls.

Figure 7: *Decision matrix created for choosing the appropriate process for the solids processing*

Declaration of AI Usage: DeepSeek and ChatGPT were both used to help format this report in L^AT_EX and in the editing process to improve the grammar and structure of this report.

References

- [1] S.-B. Kim, S. Paudel, and G. T. Seo, "Forward osmosis membrane filtration for microalgae harvesting cultivated in sewage effluent," *Environmental Engineering Research*, vol. 20, no. 1, pp. 99–104, 2015.
- [2] D. Guan, J. Zhao, Y. Wang, Z. Fu, D. Zhang, H. Zhang, and D. Wang, "A critical review on sustainable management and resource utilization of digestate," *Process Safety and Environmental Protection*, vol. 183, pp. 339–354, 2024.
- [3] W. Wang, J.-S. Chang, and D.-J. Lee, "Anaerobic digestate valorization beyond agricultural application: Current status and prospects," *Bioresour. Technol.*, vol. 373, p. 128742, 2023.
- [4] T. Wang, Y. Xue, M. Zhou, Y. Yuan, X. Zhan, S. Hu, and H. Gong, "Post-treatment of food waste digestate towards land application: A review," *Bioresource Technology*, vol. 344, p. 126211, 2022.
- [5] M. Fechter and M. Kraume, "Digestate Treatment Techniques," *Technical Transactions*, vol. 2016, no. Mechanics Issue 1-M (1) 2016, pp. 95–106, 2016.
- [6] K. Chojnacka and K. Moustakas, "Anaerobic digestate management for carbon neutrality and fertilizer use: A review of current practices and future opportunities," *Biomass and Bioenergy*, vol. 180, p. 106991, 2024.
- [7] B. Lamolinara, A. Perez, E. Yordi, C. Fiallos, K. Diéguez-Santana, and G. Ruiz-Mercado, "Anaerobic digestate management, environmental impacts, and techno-economic challenges," *Waste Management*, vol. 140, pp. 14–30, 2022.
- [8] J. J. Milledge, B. V. Nielsen, S. Maneein, and P. J. Harvey, "A brief review of anaerobic digestion of algae for bioenergy," *Energies*, vol. 12, no. 6, p. 1166, 2019.
- [9] K. Gautam, A. Pareek, and D. K. Sharma, "Exploiting microalgae and macroalgae for production of biofuels and biosequestration of carbon dioxide—a review," *International Journal of Green Energy*, vol. 12, no. 11, pp. 1122–1143, 2015.
- [10] U. B. Singh and A. S. Ahluwalia, "Microalgae: a promising tool for carbon sequestration," *Mitigation and Adaptation Strategies for Global Change*, vol. 18, no. 1, pp. 73–95, 2013.
- [11] T.-S. Sim, A. Goh, and E. W. Becker, "Comparison of centrifugation, dissolved air flotation and drum filtration techniques for harvesting sewage-grown algae," *Biomass*, vol. 16, no. 1, pp. 51–62, 1988.
- [12] K. Arola, B. Van Der Bruggen, M. Mänttäri, and M. Kallioinen, "Treatment options for nanofiltration and reverse osmosis concentrates from municipal wastewater treatment: A review," *Critical Reviews in Environmental Science and Technology*, vol. 49, no. 22, pp. 2049–2116, 2019.
- [13] C. R. Bartels, M. Wilf, K. Andes, and J. Iong, "Design considerations for wastewater treatment by reverse osmosis," *Water Science and Technology*, vol. 51, no. 6-7, pp. 473–482, 2005.
- [14] H. N. Sandefur, M. Asgharpour, J. Mariott, E. Gottberg, J. Vaden, M. Matlock, and J. Hestekin, "Recovery of nutrients from swine wastewater using ultrafiltration: Applications for microalgae cultivation in photobioreactors," *Ecological Engineering*, vol. 94, pp. 75–81, 2016.
- [15] V. V. Phat, T. Thi Thai Hang, N. C. Thuan, N. A. Dao, D. Van Tri, T. Le Luu, and C. X. Quang, "Treatment of seafood processing wastewater using an anaerobic membrane bioreactor (anmbr) for water reuse: A comparison between flat sheet and hollow fiber ultrafiltration membranes," *Journal of Environmental Chemical Engineering*, vol. 13, no. 2, p. 116066, 2025.
- [16] A. Murić, I. Petrinić, and M. L. Christensen, "Comparison of ceramic and polymeric ultrafiltration membranes for treating wastewater

from metalworking industry,” *Chemical Engineering Journal*, vol. 255, pp. 403–410, 2014.

- [17] H. Rawindran, N. Arif Bin Hut, D. K. Vrasna, P. S. Goh, J. W. Lim, C. S. Liew, and K. S. Khoo, “Ultrafiltration membrane fabricated from polyethylene terephthalate plastic waste for treating microalgal wastewater and reusing for microalgal cultivation,” *Chemosphere*, vol. 346, p. 140591, 2024.
- [18] N. Marti, A. Bouzas, A. Seco, and J. Ferrer, “Struvite precipitation assessment in anaerobic digestion processes,” *Chemical Engineering Journal*, vol. 141, no. 1-3, pp. 67–74, 2008.
- [19] A. Siciliano, C. Limonti, G. M. Curcio, and R. Molinari, “Advances in struvite precipitation technologies for nutrients removal and recovery from aqueous waste and wastewater,” *Sustainability*, vol. 12, no. 18, p. 7538, 2020.
- [20] M. I. H. Bhuiyan, D. S. Mavinic, and R. D. Beckie, “Nucleation and growth kinetics of struvite in a fluidized bed reactor,” *Journal of Crystal Growth*, vol. 310, no. 6, pp. 1187–1194, 2008.
- [21] J. Kim, C.-H. Lee, and K.-H. Choo, “Control of struvite precipitation by selective removal of nh_4^+ with dialyzer/zeolite in an anaerobic membrane bioreactor,” *Applied Microbiology and Biotechnology*, vol. 75, no. 1, pp. 187–193, 2007.
- [22] J. A. Wilsenach, C. A. H. Schuurbijs, and M. C. M. Van Loosdrecht, “Phosphate and potassium recovery from source separated urine through struvite precipitation,” *Water Research*, vol. 41, no. 2, pp. 458–466, 2007.
- [23] T.-H. Ha, N. N. N. Mahasti, H.-Q. Ha, P.-L. Liao, Y.-H. Huang, and M.-C. Lu, “Recovery of nitrogen as struvite from swine wastewater: Comparison study of batch and continuous fluidized-bed crystallization process,” *Separation and Purification Technology*, vol. 351, p. 128045, 2024.
- [24] L. Pastor, D. Mangin, R. Barat, and A. Seco, “A pilot-scale study of struvite precipitation in a stirred tank reactor: Conditions influencing the process,” *Bioresource Technology*, vol. 99, no. 14, pp. 6285–6291, 2008.
- [25] S. Shim, S. Won, A. Reza, S. Kim, N. Ahmed, and C. Ra, “Design and optimization of fluidized bed reactor operating conditions for struvite recovery process from swine wastewater,” *Processes*, vol. 8, no. 4, p. 422, 2020.
- [26] F. Tambone, B. Scaglia, G. D’Imporzano, A. Schievano, V. Orzi, S. Salati, and F. Adani, “Assessing amendment and fertilizing properties of digestates from anaerobic digestion through a comparative study with digested sludge and compost,” *Chemosphere*, vol. 81, no. 5, pp. 577–583, 2010.
- [27] B. Drosig, W. Fuchs, T. Al Seadi, M. Madsen, and B. Linke, “Nutrient recovery by biogas digestate processing,” tech. rep., IEA Bioenergy, 2015.
- [28] E. Tampio, S. Marttinen, and J. Rintala, “Liquid fertilizer products from anaerobic digestion of food waste: mass, nutrient and energy balance of four digestate liquid treatment systems,” *Journal of Cleaner Production*, vol. 125, pp. 22–32, 2016.
- [29] C. Teglia, A. Tremier, and J. L. Martel, “Characterization of solid digestates: Part 1, review of existing indicators to assess solid digestates agricultural use,” *Waste and Biomass Valorization*, vol. 2, no. 1, pp. 43–58, 2011.
- [30] R. Nkoa, “Agricultural benefits and environmental risks of soil fertilization with anaerobic digestates: a review,” *Agronomy for Sustainable Development*, vol. 34, no. 2, pp. 473–492, 2014.
- [31] J. B. Holm-Nielsen, T. Al Seadi, and P. Oleskowicz-Popiel, “The future of anaerobic digestion and biogas utilization,” *Bioresource Technology*, vol. 100, no. 22, pp. 5478–5484, 2009.
- [32] C. Vaneeckhaute, V. Lebuf, E. Michels, E. Belia, P. A. Vanrolleghem, F. M. Tack, and E. Meers, “Nutrient recovery from digestate: systematic technology review and product classification,” *Waste and Biomass Valorization*, vol. 8, no. 1, pp. 21–40, 2017.

- [33] W. Peng and A. Pivato, "Sustainable management of digestate from the organic fraction of municipal solid waste and food waste under the concepts of back to earth alternatives and circular economy," *Waste and Biomass Valorization*, vol. 9, no. 8, pp. 1255–1267, 2018.
- [34] J. Dahlin, C. Herbes, and M. Nelles, "Biogas digestate marketing: Qualitative insights into the supply side," *Resources, Conservation and Recycling*, vol. 104, pp. 152–161, 2015.
- [35] M. P. Bernal, J. A. Albuquerque, and R. Moral, "Composting of animal manures and chemical criteria for compost maturity assessment. a review," *Bioresource Technology*, vol. 100, no. 22, pp. 5444–5453, 2009.
- [36] M. Kratzeisen, N. Starcevic, M. Martinov, C. Maurer, and J. Müller, "Applicability of biogas digestate as solid fuel," *Fuel*, vol. 89, no. 9, pp. 2544–2548, 2010.
- [37] C. Y. Hung, W. T. Tsai, J. W. Chen, Y. Q. Lin, and Y. M. Chang, "Characterization of biochar prepared from biogas digestate," *Waste Management*, vol. 106, pp. 195–200, 2020.
- [38] J. Lehmann and S. Joseph, *Biochar for Environmental Management: Science, Technology, and Implementation*. London: Routledge, 2015.
- [39] S. P. Sohi, E. Krull, E. Lopez-Capel, and R. Bol, "A review of biochar and its use and function in soil," *Advances in Agronomy*, vol. 105, pp. 47–82, 2010.
- [40] D. Woolf, J. E. Amonette, F. A. Street-Perrott, J. Lehmann, and S. Joseph, "Sustainable biochar to mitigate global climate change," *Nature Communications*, vol. 1, no. 5, pp. 1–9, 2010.
- [41] P. McKendry, "Energy production from biomass (part 3): Gasification technologies," *Bioresource Technology*, vol. 83, no. 1, pp. 55–63, 2002.
- [42] A. V. Bridgwater, "Renewable fuels and chemicals by thermal processing of biomass," *Chemical Engineering Journal*, vol. 91, no. 2-3, pp. 87–102, 2003.
- [43] A. V. Bridgwater, "Review of fast pyrolysis of biomass and product upgrading," *Biomass and Bioenergy*, vol. 38, pp. 68–94, 2012.
- [44] G. W. Huber, S. Iborra, and A. Corma, "Synthesis of transportation fuels from biomass: Chemistry, catalysts, and engineering," *Chemical Reviews*, vol. 106, no. 9, pp. 4044–4098, 2006.
- [45] D. Mohan, C. U. Pittman, and P. H. Steele, "Pyrolysis of wood/biomass for bio-oil: A critical review," *Energy & Fuels*, vol. 20, no. 3, pp. 848–889, 2006.
- [46] A. Funke and F. Ziegler, "Hydrothermal carbonization of biomass: A summary and discussion of chemical mechanisms for process engineering," *Biofuels, Bioproducts and Biorefining*, vol. 4, no. 2, pp. 160–177, 2010.
- [47] J. D. Marin-Batista, J. A. Villamil, J. J. Rodriguez, A. F. Mohedano, and M. A. De la Rubia, "Valorization of digestate from biowaste through hydrothermal carbonization and anaerobic digestion," *Biomass and Bioenergy*, vol. 140, p. 105661, 2020.
- [48] A. Pantaleo, B. De Gennaro, and N. Shah, "Assessment of optimal size of anaerobic codigestion plants: An application to cattle farms in the province of bari (italy)," *Renewable and Sustainable Energy Reviews*, vol. 20, pp. 57–70, 2013.
- [49] C. A. Velis, P. J. Longhurst, G. H. Drew, R. Smith, and S. J. Pollard, "Biodrying for mechanical–biological treatment of wastes: A review of process science and engineering," *Bioresource Technology*, vol. 100, no. 11, pp. 2747–2761, 2009.
- [50] S. Guercini, G. Castelli, and C. Rumor, "Vacuum evaporation treatment of digestate: Full exploitation of cogeneration heat to process the whole digestate production," *Water Science and Technology*, vol. 70, no. 3, pp. 479–485, 2014.
- [51] M. Cerrillo, M. Moreno, L. Burgos, R. Estéfano, D. Coll, J. Soralue, N. Navarro, P. A. Arnau, and A. Bonmatí, "Low-Temperature Vacuum Evaporation of Ammonia from Pig

- Slurry at Laboratory and Pilot-Plant Scale,” *Processes*, vol. 11, p. 2910, Oct. 2023. Number: 10 Publisher: Multidisciplinary Digital Publishing Institute.
- [52] K. Dziedzic, B. Łapczyńska Kordon, M. Jurczyk, M. Arczewska, M. Wróbel, M. Jewiarz, K. Mudryk, and T. Pająk, “Solid digestate—physicochemical and thermal study,” *Energies*, vol. 14, p. 7224, 2021.
- [53] R. Picchio, F. Latterini, R. Venanzi, W. Stefanoni, A. Suardi, D. Tocci, and L. Pari, “Pellet production from woody and non-woody feedstocks: A review on biomass quality evaluation,” *Energies*, vol. 13, p. 2937, 2020.
- [54] USEPA, “Digestate and its nutrients,” 2021.
- [55] A. Slepetiene, J. Ceseviciene, K. Amaleviciute-Volunge, A. Mankeviciene, I. Parasotas, A. Skersiene, L. Jurgutis, J. Volungevicius, D. Veteikis, and I. Mockeviciene, “Solid and liquid phases of anaerobic digestate for sustainable use of agricultural soil,” *Sustainability*, vol. 15, p. 1345, 2023.
- [56] F. Tambone, V. Orzi, G. D’Imporzano, and F. Adani, “Solid and liquid fractionation of digestate: Mass balance, chemical characterization, and agronomic and environmental value,” *Bioresource Technology*, vol. 243, pp. 1251–1256, 2017.
- [57] C. Yansheng, Z. Fengliang, Z. Zhongyi, Z. Tongbin, and X. Huayun, “Biotic and abiotic nitrogen immobilization in soil incorporated with crop residue,” *Soil and Tillage Research*, vol. 202, p. 104664, 2020.
- [58] W. L. Chiyoka, F. Zvomuya, and X. Hao, “A bioassay of nitrogen availability in soils amended with solid digestate from anaerobically digested beef cattle feedlot manure,” *Soil Science Society of America Journal*, vol. 78, pp. 1291–1300, 2014.
- [59] G. Badagliacca, B. Petrovičová, S. I. Pathan, A. Roccotelli, M. Romeo, M. Monti, and A. Gelsomino, “Use of solid anaerobic digestate and no-tillage practice for restoring the fertility status of two mediterranean orchard soils with contrasting properties,” *Agriculture, Ecosystems & Environment*, vol. 300, p. 107010, 2020.
- [60] A. Crolla, C. Kinsley, and E. Pattey, “13 - land application of digestate,” in *The Biogas Handbook* (A. Wellinger, J. Murphy, and D. Baxter, eds.), pp. 302–325, Woodhead Publishing, 2013.
- [61] S. Higgins, K. Schmidt, and A. Gumbert, “Kentucky nutrient management planning guidelines (kynmp),” 2024.
- [62] B. Lamolinara, A. Perez, E. Yordi, C. Fiallos, K. Diéguez-Santana, and G. Ruiz-Mercado, “Anaerobic digestate management, environmental impacts, and techno-economic challenges,” *Waste Manag.*, vol. 140, pp. 14–30, 2022.
- [63] R. O’Shea, R. Lin, D. M. Wall, J. D. Browne, and J. D. Murphy, “A comparison of digestate management options at a large anaerobic digestion plant,” *Journal of Environmental Management*, vol. 317, p. 115312, 2022.